DONGWOOK KIM

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Interest

Research interests include 2D/3D Computer Vision and Synthetic Data Generation (Digital Twin). Recently, I have been focusing on **Data-Centric AI** and **Intelligent Agent Systems** to enhance the efficiency and usability of AI models in complex industrial environments.

Education

UNIST. Graduate School of Artificial Intelligence, AI core — *Ulsan, S.Korea*
Jan 2024 – Current

- Total GPA 4.2/4.3
- Visual Information Processing Lab (Prof.Jae-Young Sim)
- Research Focus: Dataset Distillation, AI-Agent, and 3D Gaussian Splatting.

Kwangwoon Univ. B.S in Information Convergence, Major in Data Science — *Seoul, S.Korea*
Jan 2018 – 2024

- Total GPA 3.98/4.5, Major GPA 4.16/4.5
- Deep Imaging and Graphics Lab (Prof.Dongjoon Kim)
- Research Focus: 3D Scene Reconstruction(NeRF) and LiDAR Signal Processing.

Experience

Qualcomm Institute, UC San Diego — *San Diego, CA, USA*
AI Development Project Intern — *Jul. 2022 – Aug. 2022*

- Analyzed user patterns to classify and detect abusive behavior.
- Collaborated with international teams, communicating research progress in English.

Publications

International

ICLR 2026 — Apr 2026
Parameterization-Based Dataset Distillation of 3D Point Clouds through Learnable Shape Morphing — 1st author, link
Dongwook Kim, Jae-Young Yim* and Jae-Young Sim*

NeurIPS 2025 — Dec 2025
Dataset Distillation of 3D Point Clouds via Distribution Matching — 1st author, link
Jae-Young Yim, Dongwook Kim* and Jae-Young Sim*

ICIP-W 2025 — Sep 2025
Class-Aware Coreset Selection for 3D Point Clouds Classification — 2nd author
Jae-Young Yim, Dongwook Kim and Jae-Young Sim

Domestic

IPIU 2024 — Feb 2025
Controllable Classification via Negative-Context-Aware Learning — 2nd author, link
YoonHan Kim, Joohyeok Kim, Dongwook Kim and Jae-Young Sim

IEIE 2024 — Jun 2024
Enhancing Quality of Gaussian Splatting in Few Shot Condition by Multi-Scale Augmentation — 1st author, link
Dongwook Kim, Jae-Young Yim and Jae-Young Sim

HCI Academy of Korea 2023 Feb 2023
Synthesized Training Data for a Ship 3D Surround View Learning Model Based on User Evaluations 1st author, link
Dongwook Kim, Jonghun Kim, Taemin Jeong and Dongjoon Kim

Projects

Generative AI for Multimodal Transportation Analytics (with MSIT) Nov. 2025

- Developing an explainable multimodal Decision Support Agent integrating CCTV/sensor data with an LLM-RAG pipeline.
- Building robust vision models for adverse-weather traffic scenes and constructing synthetic datasets via Unreal Engine simulation.
- Designing a multimodal traffic knowledge graph and spatio-temporal GNNs for traffic prediction and anomaly detection.

Distortion-Free Surround View Monitor System (with HD Hyundai) Jun. 2023
link

- Built a high-fidelity Digital Twin using Unreal Engine 5, applying camera calibration to match virtual sensors with real-world optics.
- Developed End-to-End deep learning pipelines for sensor fusion (LiDAR + RGB) and super-resolution.
- Achieved Simulation-to-Real adaptation for autonomous navigation support.

Time-Series Anomaly Detection & Signal Processing Jul. 2022

- Developed an ensemble ML pipeline to detect anomalies in tabular data and multi-channel sensor signals.
- Applied DL-based signal classification and feature engineering for multimodal time-series data.

Competition & Awards

2nd Prize, 2025 Samsung AI Challenge Oct. 2025
link

- Developed an AI co-scientist agent integrating MCP and RAG architectures.
- Implemented dynamic context management using ChromaDB to ground dialogues in technical documents.

1st Prize, SKT FLY AI Competition Feb. 2024
link

- Built a multimodal Task-Oriented Agent integrating visual signals and conversational context.
- Designed a Context-aware conditioning pipeline using text embeddings to guide reliable response generation and action selection.

1st Prize (Minister's Award), AI Contest for Software-Centered Universities Oct. 2022

- Competed in an OCR task for signage image recognition.

2nd Prize, Student Creative Design Competition Jun. 2022

- Developed a StyleGAN-based generative model for book cover synthesis under limited labeled data conditions.
- Led the End-to-End development process, including AI modeling, web backend, and mobile deployment.

Patents

Distribution-Matching-Based Dataset Distillation Method for 3D Point Clouds
KR Patent Application No. 10-2025-0214325 (2025; international filings planned)

Technologies

Languages: Python, C++, JavaScript
Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, Open3D,LangGraph, LangChain, MCP, RAG,
AI & 3D Vision: Dataset Distillation, LLMs,multi-agent orchestration, 3D Scene Reconstruction(NeRF/GaussianSplatting), Sensor/Camera Calibration, XR
Simulation & Digital Twin: Unreal Engine 5, Three.js, Panda3D
MLOps & Tools: Docker, Git/GitHub